Computer Networks

LECTURE-4

Recap of Lecture 3

- Physical Structure of Network
  - Physical Topologies
    1. Bus
    2. Mesh
    3. Star
    4. Ring
    5. Tree
    6. Hybrid

Overview of Lecture 4

- Categories of Networks
  - PAN
  - LAN
  - MAN
  - WAN
  - Interconnection of Networks: Internetworks
  - Internet

Categories/ types of Networks

Categories of Networks

- Computer networks can be characterized by their;
  1. Size
  2. purpose.

Categories of Networks (by size)

- The size of a network can be expressed by the geographic area they occupy and the number of computers that are part of the network.

- Networks can cover anything from a handful of devices within a single room to millions of devices spread across the entire globe.

- Some of the different networks based on size are:
  1. Personal area network (PAN)
  2. Local area network (LAN)
  3. Metropolitan area network (MAN)
  4. Wide area network (WAN)

Categories of Networks (by purpose)

- In terms of **purpose**, many networks can be considered general purpose, which means they are used for everything from sending files to a printer to accessing the Internet.

- Some types of networks, however, serve a very particular purpose.

- Some of the different networks based on their main purpose are:
  1. Storage area network, or SAN
  2. Enterprise private network, or EPN
  3. Virtual private network, or VPN

Personal Area Network

- Network organized around an individual person within a single building (small office or residence).

- A typical PAN would include one or more
  - computers,
  - telephones,
  - peripheral devices,
  - video game consoles
  - and other personal entertainment devices.

- If multiple individuals use the same network within a residence, the network is sometimes referred to as a home area network, or HAN.

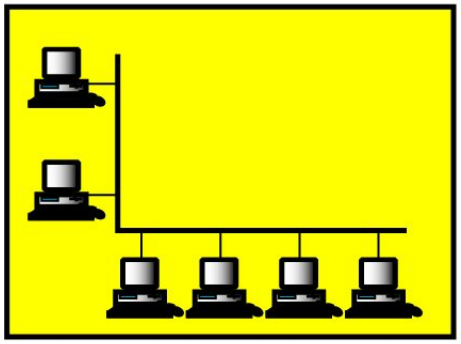
Personal Area Network (cont..)

- In a very typical setup, a residence will have a single wired Internet connection connected to a modem.
- This modem then provides both wired and wireless connections for multiple devices.
- The network is typically managed from a single computer but can be accessed from any device.
- This type of network provides great flexibility.
- For example, it allows you to:
- Send a document to the printer in the office upstairs while you are sitting on the couch with your laptop.
- Upload the photo from your cell phone to your desktop computer.
- Watch movies from an online streaming service to your TV.
- If this sounds familiar to you, you likely have a PAN in your house without having called it by its name.

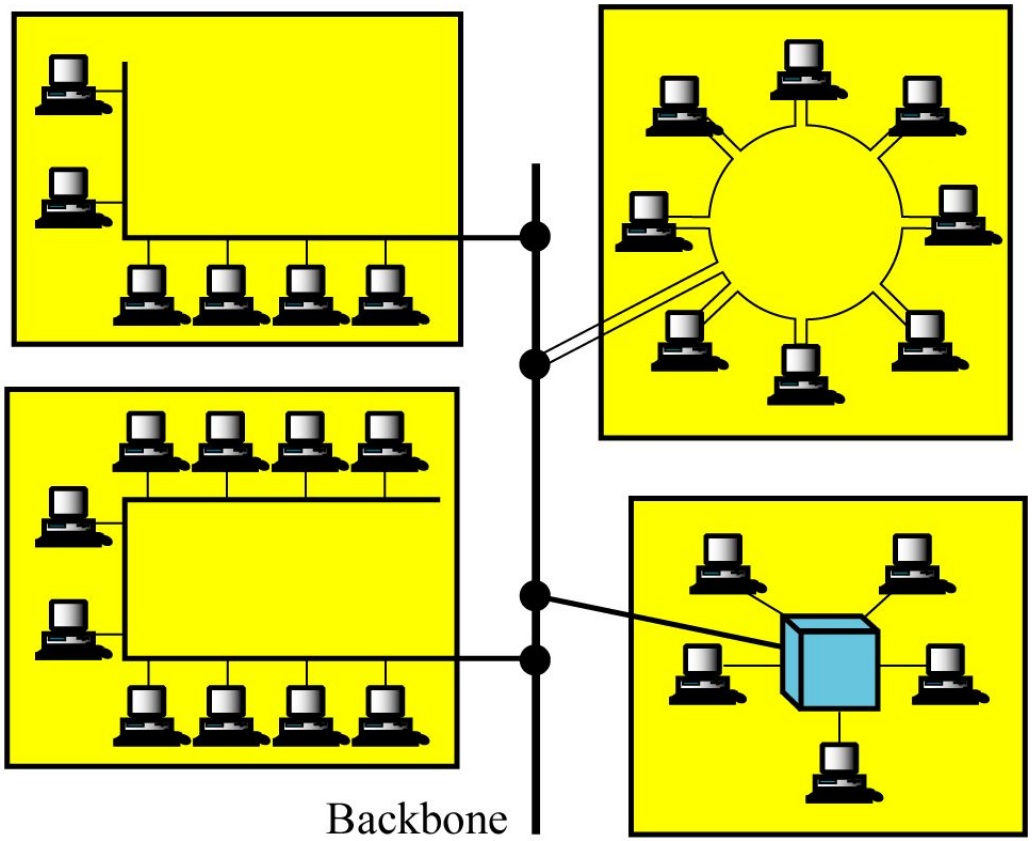
Local Area Networks

"A local area network (LAN) is a data communication system that allows a number of independent devices to communicate directly with each other in a limited geographic area."

Local Area Networks (cont..)



a. Single building LAN



b. Multiple building LAN

Local Area Networks (cont..)

- Privately owned and links the devices within a small geographical area.
- LAN is basically established in a
  - single office,
  - building,
  - campus,
  - or a city.
- Depending on the needs of an organization and the type of technology used, a LAN can as simple on two PCs and printer or it can extend throughout a company and include voice, sound and video peripherals.
- Currently, LAN size is limited to a few kilometers.

Local Area Networks (cont..)

- Share Resources between PC or Workstation.
- Resources to be shared are
  - Hardware
    - Printer, Scanner, CD-Rom
  - Software
    - an application program
    - or data.
- LANs are distinguished from other types of network by
  - Geographical Area (Size)
  - transmission media
  - topology.

Local Area Networks (cont..)

- In general, a given LAN will use only one type of transmission medium.
- The most common LAN topologies are
  - Bus,
  - Star
  - Ring.
- Traditionally, LANs have data rates in the
  - 4Mbps
  - 16Mbps
  - 100Mbps to 1000Mbps with gigabits systems in development.

LAN Access Methods

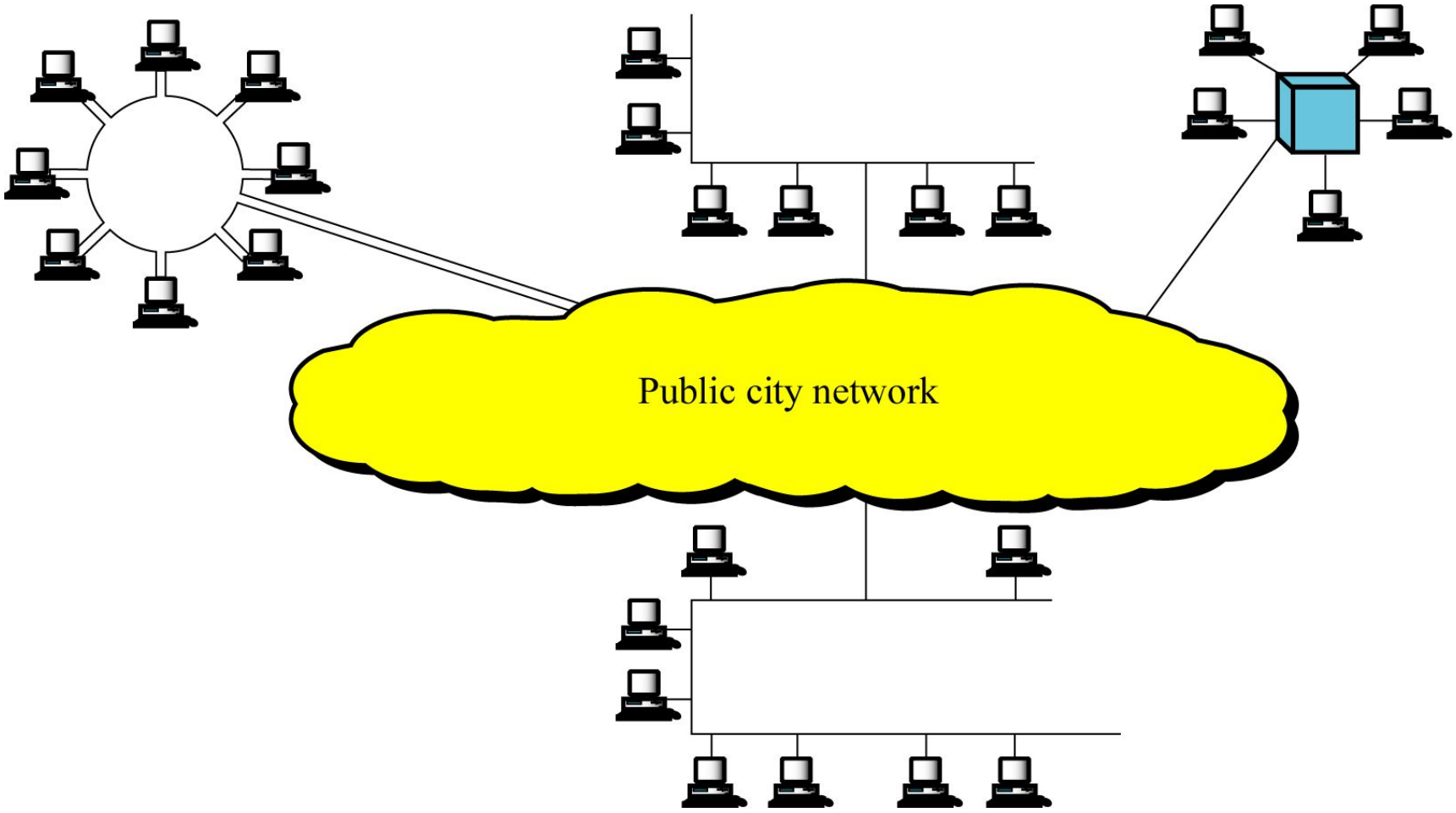
- **Broadcasting**
  - In a broadcast LAN, transmitted information will be received by all stations simultaneously. The medium access schemes are random access such as CSMA/CD which may cause contention, and controlled access such as token-passing, in which no contention will occur.

- **Switching**
  - In a switched architecture, a switch forward data packets to their destinations that may be a single user station or another LAN segment.

Various Local Area Networks

- Ethernet
  - Fast Ethernet,
  - Gigabit Ethernet
  - 10G Ethernet
- Fiber Channel
- Hipper LAN
- Token ring
- ATM LAN
- FDDI (Fiber Distributed Data Interface)
- Wireless LAN
- ……..
- There are also some other technologies such as 100VG, token bus, ARCnet, but those are almost obsolete.

Metropolitan Area Networks



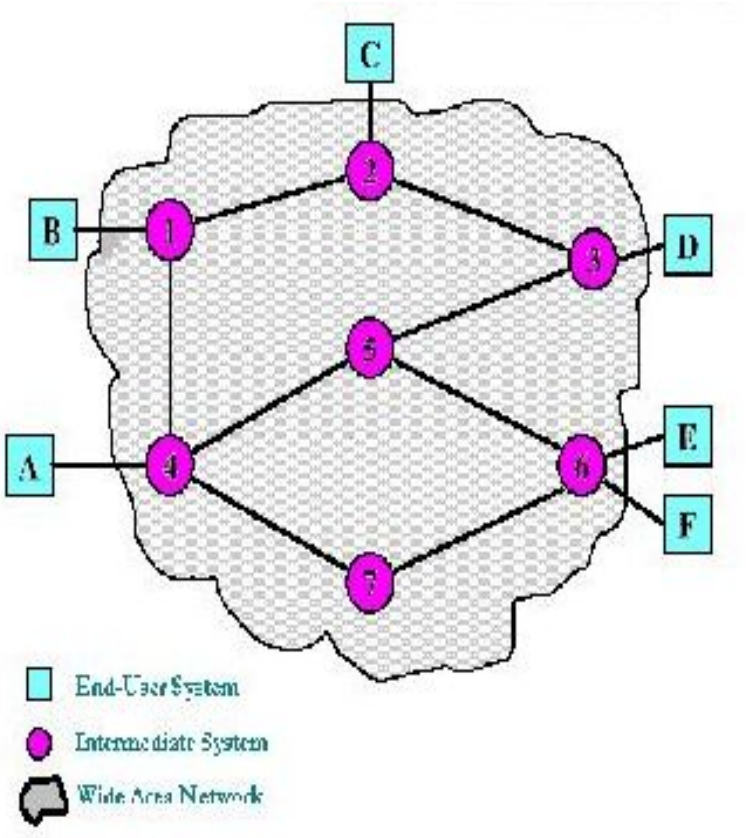
Metropolitan Area Networks (MANs)

- A MAN is designed to extend over an entire city or state.
- It may be a single network such as a cable television network, or it may be a means of connecting a number of LANs into a large network so that resources may be shared LAN-to-LAN as well as device-to-device.
  - For example, a company can use a MAN to connect the LANs in all of its offices throughout a city.
- A MAN may be wholly owned and operated by a private company, or it may be a service provided by a public company such as local telephone company. A MAN is larger than LAN but smaller than WAN in size.

Wide Area Networks

- Networking implemented on far away or long geographical distances.
  - For Example, networking implemented between two continents or countries.
- WAN is always based on connection between different LANs and MANs.
- For Example, network between campuses of a university in different states of a country.

Wide Area Networks





Wide Area Networks

"WAN provides large distance transmission of data, voice, image and video information over large geographical areas that may comprise a country, a continent or even the whole world."

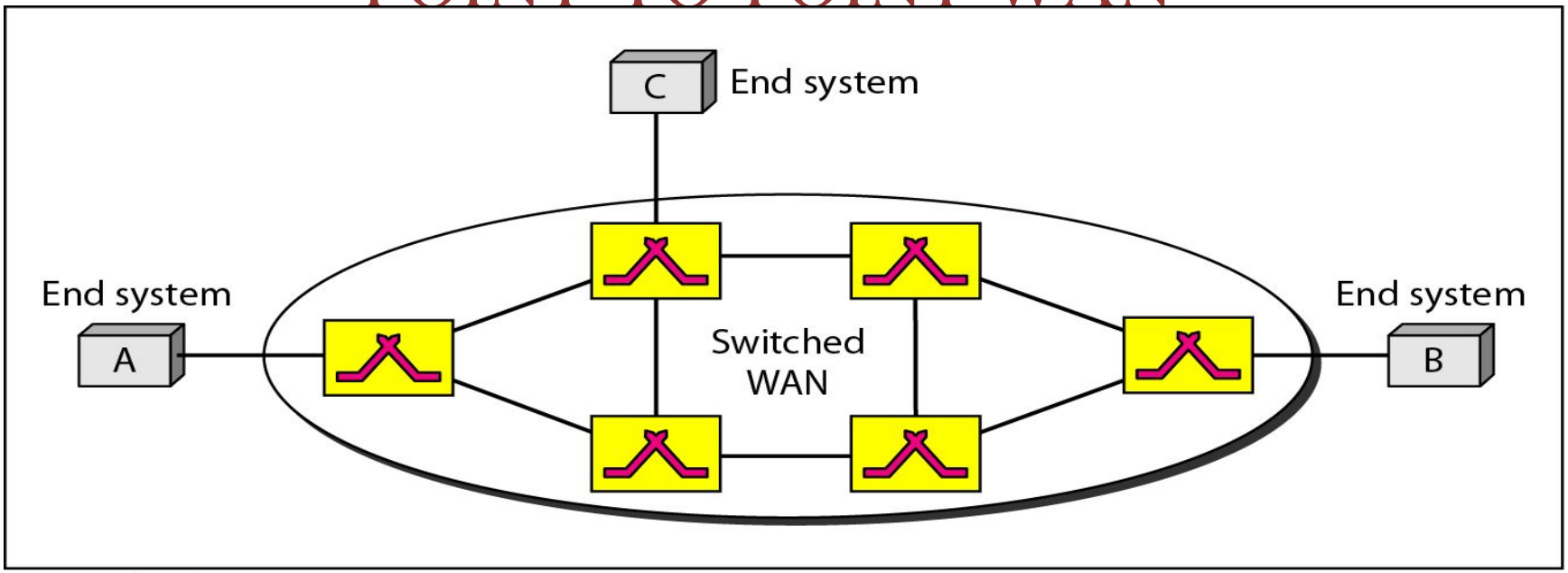
Wide Area Networks

- **TYPES OF WAN**
  - Enterprise WAN
    - A WAN that is wholly owned and is used by a single company is known as “Enterprise WAN”.
    - In other words network between two or more cities, states, or countries for a specific person or company,
    - e.g. Network of Habib Bank Limited in Pakistan, and network of NADRA throughout Pakistan.
  - Global WAN
    - A WAN that is accessible throughout the world for anybody is known as Global WAN,
    - e.g. Internet and PSTN all over the world.

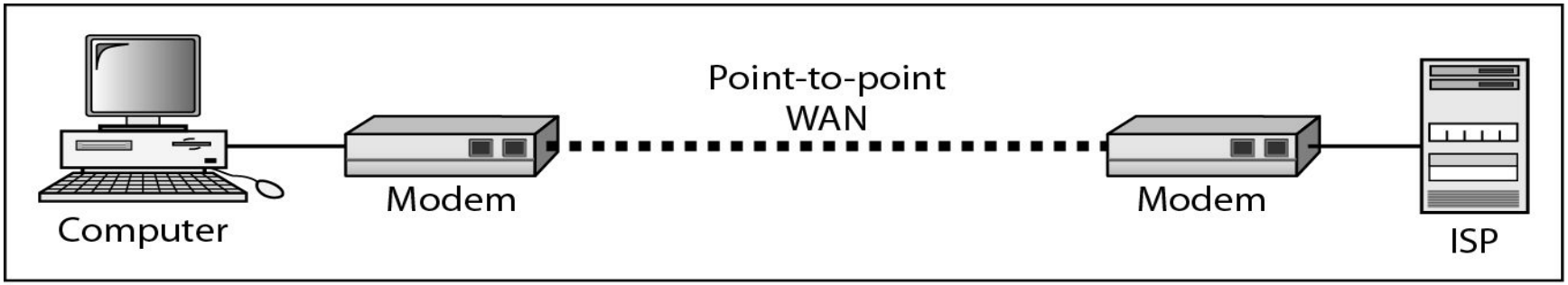
Wide Area Networks

- A WAN can be as complex as the backbones that connect the Internet or as simple as a dial-up line that connects a home computer to the Internet.
- We normally refer to the
  - first as a switched WAN
  - and to the second as a point-to-point WAN.
- **The switched WAN** connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN. Examples are:
  - X.25
  - Frame Relay.
  - Asynchronous Transfer Mode (ATM) network
  - wireless WAN.
- **Point-to-point WAN** is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP). This type of WAN is often used to provide Internet access.

WANS: A SWITCHED WAN AND A POINT-TO-POINT WAN



a. Switched WAN

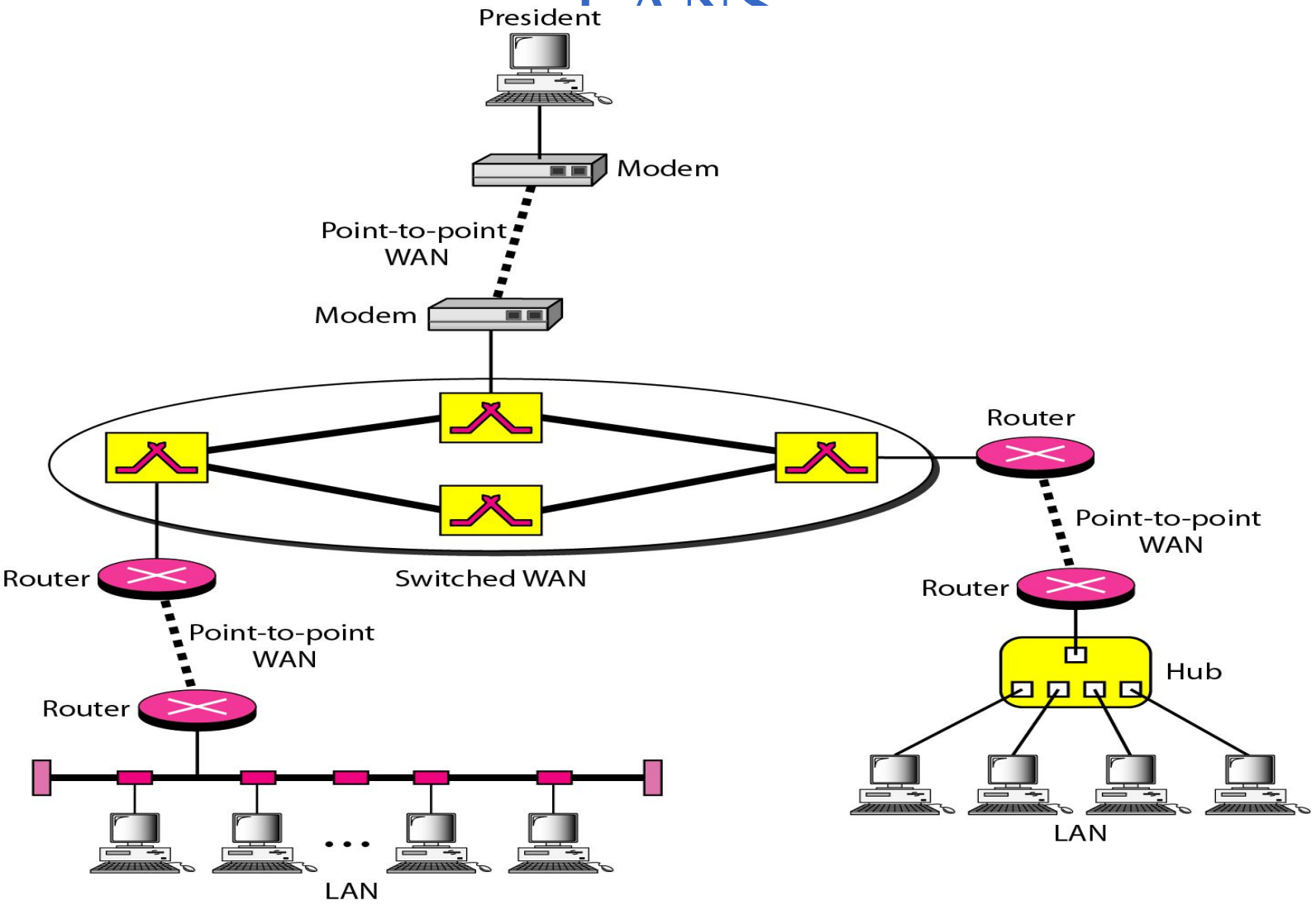


b. Point-to-point WAN

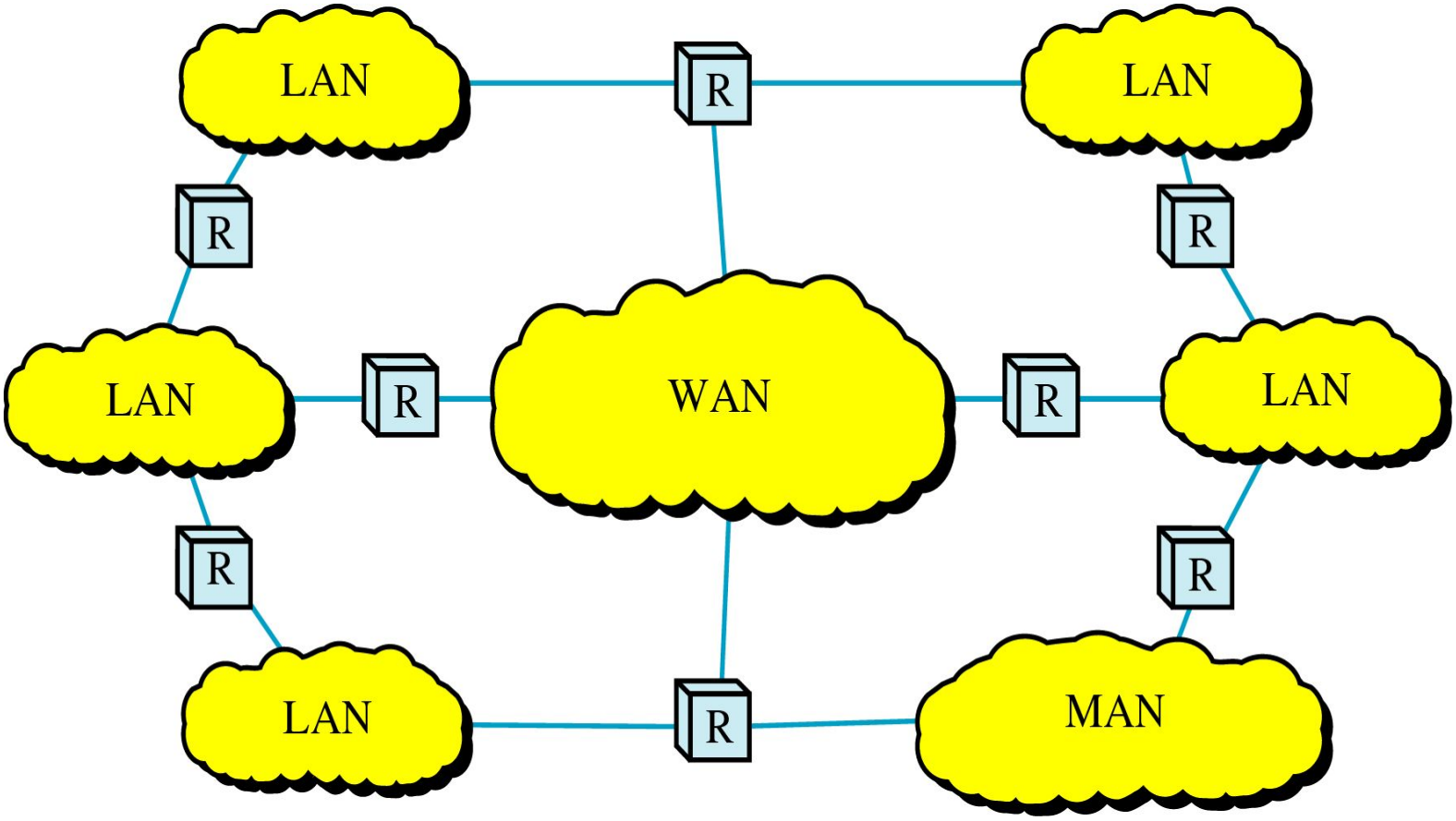
Internetworks

- When two or more networks are connected, they become an internetwork, or internet.
- As an example, assume that an organization has two offices, one on the east coast and the other on the west coast. The established office on the west coast has a bus topology LAN; the newly opened office on the east coast has a star topology LAN.
- The president of the company lives somewhere in the middle and needs to have control over the company from his house.
- To create a backbone WAN for connecting these three entities (two LANs and the president's computer), a switched WAN (operated by a service provider such as a telecom company) has been leased.
- To connect the LANs to this switched WAN, however, three point-to-point WANs are required.
- These point-to-point WANs can be a high-speed DSL line offered by a telephone company or a cable modern line offered by a cable TV provider.

A HETEROGENEOUS NETWORK MADE OF FOUR WANS AND TWO LANS



Internetworks



Summary

- Categories of Networks
  - PAN
  - LAN
  - MAN
  - WAN
  - Interconnection of Networks: Internetworks
  - Internet

ASSIGNMENT # 1

“Brief Introduction and History of INTERNET”

Suggested Reading

- Section
  - 1.2,
  - 1.3,
    - “Data Communications and Networking” 4th Edition by Behrouz A. Forouzan
- Sections
  - 1.3,
    - “Data and Computer Communication” 6th Edition by William Stallings